

RetinalAI: Hackathon & Pitch Guide

Classification: Presentation Playbook **Version:** 2.1 | April 2026 **Purpose:** Step-by-step guide for presenting RetinalAI at hackathons, investor pitches, and demo days **Focus Market:** Uganda and East Africa — grounded in real healthcare data

Uganda Healthcare Reality (Verified Data, 2025-2026)

These numbers anchor every pitch. Memorize them — they are your ammunition.

Fact	Number	Source
Uganda population	~48 million (2026 est.)	UBOS
Ophthalmologists in Uganda	~60 (half based in Kampala)	PMC / Sightsavers 2025
Ophthalmologist-to-population ratio	1 per 800,000	Sightsavers Eye Health Assessment 2025
WHO recommended ratio	4 per 1,000,000	WHO
Ugandans with vision impairment/blindness	1.4 million+	Uganda National Eye Health Survey
Ugandans living in rural areas	84%	Wikipedia / UBOS
Diabetic retinopathy prevalence (Mulago study)	19.5% of diabetic patients	Mulago Hospital DR Study, PubMed
Sight-threatening DR (of all DR cases at Mulago)	85.7%	PubMed 31063850
DR prevalence in East Africa (meta-analysis)	~33% of diabetic patients	PLOS ONE 2024
Vision impairment among diabetics in SSA	29% (7% blind)	PLOS ONE 2025
Uganda health budget FY 2025/26	US\$ 5.87 trillion (~\$1.5B, 8.1% of budget)	UNICEF Uganda Budget Brief
Per capita health spending	~\$23/year (vs WHO target \$86)	P4H Network
Health centers within 5km of population	35%	PMC Equity Study
Mulago Hospital ophthalmology	Largest eye department in Uganda, teaching hospital for Makerere University	Mulago NRH
Uganda Digital Health Strategy	National Health Information & Digital Health Strategic Plan 2023-2025 launched	MoH Uganda
Rocket Health (Uganda mHealth)	25,000 patients/month, \$5M Series A	ICTworks

The 5 Questions Every Pitch Must Answer

Every pitch format in this guide is built around these five questions. Nail all five and the audience remembers you.

#	Question	RetinalAI Answer — Uganda-Grounded (30 seconds or less)
01	What is the PROBLEM?	Uganda has 60 ophthalmologists for 48 million people — half of them work in Kampala. 84% of Ugandans live in rural areas with no specialist access. At Mulago Hospital, 19.5% of diabetic patients already have retinal disease, and 85.7% of those cases are sight-threatening. These patients waited too long because there was no one to screen them. Current AI tools detect 1-3 diseases at \$25-40/test — unaffordable when health spending is \$23/person/year.
02	WHO has this problem?	Uganda's 6,000+ health centers that have zero eye screening capability. The Ministry of Health, which doubled the health budget to \$1.5B but still cannot hire enough ophthalmologists. Mulago Hospital's eye clinic, which runs two clinics but needs AI triage to handle volume. Every Health Centre III and IV across Uganda's 135 districts that sees diabetic patients but cannot screen their eyes.
03	What is your SOLUTION?	RetinalAI takes a single fundus photograph, validates it's a real retinal image, then screens for 45 eye diseases in under 200 milliseconds — at 50 cents per scan, less than UGX 2,000. It shows the clinician exactly why each disease was flagged with a heatmap, prioritizes which patients need urgent referral to Mulago or regional hospitals, and learns from every correction a doctor makes. It works offline on a laptop with no internet — critical for rural Uganda.
04	Why YOU / Why NOW?	We built this with Uganda's disease epidemiology embedded in the model — 144 disease relationships calibrated for local prevalence. Uganda just launched its National Digital Health Strategic Plan and doubled the health budget. The WHO mandated universal eye screening in 2025. We are ready to deploy at Mulago and expand to every district hospital. Our production stack — OpenTelemetry, MLflow, active learning, edge deployment — means this isn't a research demo. It's deployable infrastructure.
05	What do you WANT?	\$2-4M Seed to run clinical validation at Mulago National Referral Hospital in partnership with Makerere University's Department of Ophthalmology, deploy pilots at 5 regional referral hospitals across Uganda, and file CE Mark by Q4 2026. Three milestones before Series A: clinical study published, first MoH contract, CE Mark granted.

1. Before the Event: Preparation Checklist

Technical Setup (Do 1 Hour Before)

```
# 1. Start the full stack (backend + frontend + observability)
make up-phase1 # starts API + OTEL Collector + Jaeger + MLflow
cd frontend && bun start &

# 2. Verify backend health
curl http://localhost:8080/health

# 3. Verify frontend
open http://localhost:3000

# 4. Pre-load a test prediction to warm the model cache
curl -X POST http://localhost:8080/api/v1/predict -F "file=@test_images/sample_fundus.jpg"

# 5. Verify governance endpoints are live (impressive for live demo)
curl http://localhost:8080/api/v1/governance/drift
curl http://localhost:8080/api/v1/governance/active-learning-stats

# 6. Open Jaeger UI in a background tab (shows distributed traces live)
open http://localhost:16686

# 7. Clear browser cache, set display to 1920x1080, disable OS notifications
# 8. Keep a mobile hotspot as network backup
```

Demo Images Ready

Image	Expected Result	Purpose
Healthy fundus	Normal / No disease	Shows the system handles negatives correctly
Diabetic retinopathy (severe)	DR + related conditions, URGENT referral	The “wow” moment — multi-label + clinical reasoning
Glaucoma suspect	ODC/ODP, ROUTINE referral	Shows referral priority variation
Mixed pathology	Multiple conditions, knowledge graph adjustments visible	Demonstrates clinical knowledge graph reasoning
Non-fundus image	Rejected by fundus gate	Shows safety validation (not a retinal photo)

Demo Tabs Ready (Impressive Background)

1. **RetinalAI frontend** — the main demo
2. **Jaeger UI** — shows distributed traces flowing in real time during demo

3. **Governance API** — `GET /api/v1/governance/drift` showing live drift monitoring
4. **MLflow UI** — model registry with version history (if MLflow is running)

Backup Plan

- **If the live demo fails:** Have 2-minute screen recording of a successful demo ready
- **If network fails:** Backend and frontend both run locally in Docker — no cloud dependency
- **If projector/screen fails:** Have key screenshots on your phone for a verbal walkthrough

2. The 3-Minute Hackathon Pitch (Uganda-Grounded)

Context: Fast-paced, technical judges, 20+ teams presenting, attention spans are short. **Goal:** Make them remember you. Problem-demo-impact. Every number is real.

Structure (maps to the 5 Questions)

[0:00 - 0:25]	THE HOOK	→ Q1 PROBLEM + Q2 WHO
[0:25 - 0:50]	THE PAIN	→ Q1 PROBLEM (make them feel it)
[0:50 - 2:00]	LIVE DEMO	→ Q3 SOLUTION (show, don't tell)
[2:00 - 2:35]	WHY US, WHY NOW	→ Q4 WHY YOU / WHY NOW
[2:35 - 3:00]	IMPACT & ASK	→ Q5 WHAT DO YOU WANT

Script

THE HOOK (0:00 - 0:25) — Q1 + Q2

“Uganda has 60 ophthalmologists for 48 million people. Half of them work in Kampala. If you're a diabetic patient in Gulu, or Arua, or Kabale — the nearest eye specialist is a full day's travel away. At Mulago Hospital, a study found that 19.5% of diabetic patients already have retinal disease — and 85.7% of those cases are sight-threatening. They went blind waiting. We built RetinalAI to make sure no one has to.”

[RetinalAI dashboard is already loaded on screen]

THE PAIN (0:25 - 0:50) — Q1

“Grace is a 54-year-old diabetic woman at a Health Centre IV in Gulu. There’s no ophthalmologist in her district — the nearest is at Lacor Hospital, 340 kilometers from Kampala. An eye screening costs more than what she earns in a month. Uganda spends \$23 per person per year on healthcare — a quarter of the WHO minimum. She waits. And waits. By the time she’s seen, she has advanced glaucoma. Irreversible.”

“Existing AI tools? They detect 1-3 diseases at \$25-40 per test. That’s almost two months of Uganda’s per-capita health budget — for a single screening.”

LIVE DEMO (0:50 - 2:00) — Q3

“Let me show you what we built.”

[Navigate to Screening page]

“I’m uploading a retinal fundus image — the kind captured by a \$500 portable camera that any Health Centre III could afford. First, our safety gate — a learned MobileNetV3 fused with statistical checks — confirms this is a valid retinal photograph in under 12 milliseconds.”

[Drag-and-drop image, click ‘Analyze Image’]

“200 milliseconds. 11 conditions detected. URGENT referral to the nearest regional hospital. Look — the system didn’t just predict diseases in isolation. Our Clinical Knowledge Graph — 144 disease relationships calibrated for Uganda’s disease epidemiology — boosted the probability of Cystoid Macular Edema because it co-occurs with Diabetic Retinopathy. This is differential diagnosis, not pattern matching.”

[Click GradCAM tab]

“And we don’t ask ophthalmic clinical officers to trust a black box. GradCAM shows exactly which retinal regions drove each prediction. The OCO at a rural health centre can see the evidence, confirm or correct it, and that correction trains the model to get better.”

[Quick flash to Jaeger tab]

“Every prediction is traced end-to-end with OpenTelemetry — visible in real time. Every result logged in an immutable audit trail. This is how you build trust with the Ministry of Health.”

WHY US, WHY NOW (2:00 - 2:35) — Q4

“Under the hood: a RETFound foundation model pretrained on 1.6 million retinal images, fine-tuned with LoRA adapters. A LangGraph agentic pipeline where Claude AI reasons like a clinician — with automatic fallback to deterministic rules when there’s no internet. This runs offline on a laptop with ONNX edge deployment.”

“This is not a notebook. It’s a production system — 188 automated tests, MLflow model registry, active learning that fine-tunes when clinicians correct predictions, drift detection, immutable audit logs. Uganda just launched its National Digital Health Strategic Plan and doubled the health budget to \$1.5 billion. The WHO mandated universal eye screening in 2025. The infrastructure moment is now.”

IMPACT & ASK (2:35 - 3:00) — Q5

“One photograph. 45 diseases. 200 milliseconds. Under UGX 2,000 per scan — less than the cost of a boda-boda ride. We can screen every diabetic patient at every Health Centre IV in Uganda’s 135 districts.”

“We’re raising a Seed round to run clinical validation at Mulago Hospital with Makerere University’s Department of Ophthalmology, then deploy to five regional referral hospitals. RetinalAI. Early detection saves sight.”

[Hold. Let it land. Do not rush to Q&A.]

3. The 5-Minute Investor Pitch

Context: Investor audience, looking for market size, defensibility, and team. **Goal:** Get the follow-up meeting.

Structure (5 Questions, expanded)

```
[0:00 - 0:30] Q1 PROBLEM — HOOK + MARKET PAIN
[0:30 - 1:30] Q3 SOLUTION — LIVE DEMO (condensed)
[1:30 - 2:30] Q4 WHY US — CLINICAL KNOWLEDGE GRAPH (the moat)
[2:30 - 3:30] Q2 WHO — MARKET & BUSINESS MODEL
[3:30 - 4:30] Q4 WHY NOW — TRACTION & REGULATORY
[4:30 - 5:00] Q5 THE ASK
```

Script

Q1: PROBLEM — HOOK + MARKET PAIN (0:00 - 0:30)

"2.2 billion people worldwide have vision impairment. There are only 232,000 ophthalmologists to serve them. The ophthalmic AI market is \$5.4 billion and growing at 31% annually. Every existing solution screens for 1-3 diseases at \$25-40 per test. We screen for 45 diseases at 50 cents."

Q3: SOLUTION — LIVE DEMO (0:30 - 1:30)

[Same demo flow as hackathon, slightly faster — upload, results, GradCAM. 60 seconds.]

"One scan. 45 diseases. 200 milliseconds. Explainable results with referral prioritization. Every prediction traced with OpenTelemetry, logged to an immutable audit trail, and monitored for drift in real time."

Q4: WHY US — CLINICAL KNOWLEDGE GRAPH (1:30 - 2:30)

"Here's why this is hard to replicate. Our Clinical Knowledge Graph encodes 144 disease relationships from peer-reviewed ophthalmology literature — co-occurrence patterns, severity hierarchies, and population-specific prevalence data calibrated for Uganda's disease epidemiology."

"When our model detects Diabetic Retinopathy with high confidence, the knowledge graph automatically adjusts related condition probabilities — just like a specialist would reason through a differential diagnosis. This isn't a model weight that any team with GPUs can reproduce — it's structured clinical knowledge."

"On top of that: a LangGraph agentic pipeline where Claude AI acts as a clinical reasoning co-pilot. An active learning loop that automatically fine-tunes the model when clinicians correct predictions. MLflow model registry with staging, shadow deployment, and promotion gates. Ray Serve with dynamic batching and canary releases. Five explainability methods. 188 automated tests. This is production infrastructure, not a demo."

Q2: WHO — MARKET & BUSINESS MODEL (2:30 - 3:30)

“Our go-to-market follows the ‘start where you’re the only option’ playbook.”

“Phase 1 — East Africa. One ophthalmologist per million people. No AI competition. We’re already calibrated for local disease prevalence. We partner with Ministries of Health at 30-50 cents per scan.”

“Phase 2 — UK and EU. CE Mark pathway is ready, and we’re already EU AI Act compliant — most competitors will need to retrofit. NHS AI Diagnostic Fund is actively seeking solutions like ours.”

“Phase 3 — US. FDA De Novo submission. No predicate exists for 45-disease screening — we become the predicate. That’s a regulatory moat.”

“Revenue model: per-scan fees for government programs, SaaS subscriptions for clinics, enterprise licenses for hospital networks, OEM royalties for camera manufacturers embedding our API. Blended gross margin: 76%.”

“Year 1 target: \$1.76M ARR. Year 3: \$25.5M. Break-even at month 18.”

Q4: WHY NOW — TRACTION & REGULATORY (3:30 - 4:30)

“Where we are today: working product with 45-disease classification, a full 2026 production backend — OpenTelemetry distributed tracing, MLflow model registry, active learning closed loop, drift detection with NannyML and Evidently, immutable Kafka audit logs, circuit breakers, graceful degradation, ONNX edge deployment. 188 automated tests. EU AI Act compliance embedded from day one.”

“Regulatory pathway: CE Mark submission Q4 2026, FDA Pre-Sub meeting Q3 2026, East African registration Q1 2027. We’ve already engaged with Mulago National Referral Hospital in Kampala for our first clinical validation site.”

“IP: First provisional patent filing Q3 2026 — clinical knowledge graph-augmented GNN for multi-label medical image classification. Three additional patents in the pipeline.”

Q5: THE ASK (4:30 - 5:00)

“We’re raising a \$2-4 million Seed round to execute clinical validation, CE Mark submission, and first commercial pilots.”

“Use of funds: 40% clinical validation and regulatory, 30% engineering and infrastructure, 20% first three hires — VP Regulatory, Clinical Director, ML Engineer — and 10% operating reserve.”

“We’ll hit three milestones before the next raise: CE Mark granted, retrospective study published, and first paying customers in East Africa and the UK.”

“We’re making AI screening possible for the 4 billion people who can’t access an eye specialist. Let’s talk.”

4. The 10-Minute Demo Day Presentation

Context: Mix of investors, partners, press, and peers. More time for depth. **Goal:** Full understanding of product, technology, market, and team.

Structure

```
[0:00 - 1:00] Q1 — PROBLEM NARRATIVE (Grace's story)
[1:00 - 3:00] Q3 — FULL LIVE DEMO (with clinical reasoning deep-dive)
[3:00 - 4:00] Q3 — EXPLAINABILITY + OBSERVABILITY SHOWCASE
[4:00 - 5:30] Q4 — ARCHITECTURE & PRODUCTION STACK
[5:30 - 7:00] Q2 — MARKET & COMPETITIVE POSITIONING
[7:00 - 8:30] Q2 — BUSINESS MODEL & FINANCIALS
[8:30 - 9:30] Q4 — REGULATORY & TRACTION
[9:30 - 10:00] Q5 — VISION & ASK
```

Key Additions (vs. 5-Minute Pitch)

Problem Narrative (Q1 — make them feel the pain):

“Grace is a 54-year-old diabetic woman in Gulu, Northern Uganda. The nearest ophthalmologist is in Kampala — 340 kilometers away. She’s been losing peripheral vision for two years, but a screening would cost more than her monthly income. By the time she saves enough to travel, she has advanced glaucoma with irreversible vision loss. This story repeats 80 million times a year across Africa.”

Architecture Deep-Dive (Q4 — show why no one else can do this):

“Four custom graph neural network architectures evaluated in rigorous cross-validation. A RETFound ViT-Large backbone pretrained on 1.6 million retinal images with LoRA adapters — only 2.4 million trainable parameters on a 304 million parameter foundation.”

“A LangGraph agentic pipeline: six processing nodes — classify, triage, reason, explain, review, report — where Claude AI acts as a clinical reasoning co-pilot with automatic fallback to Groq, then deterministic rules. The system never fails silently — it degrades gracefully through four levels.”

2026 Production Stack (Q4 — the moat that isn’t just the model):

“This isn’t a Jupyter notebook demo. OpenTelemetry distributed tracing with Jaeger — every prediction traced from HTTP request through model inference to clinical reasoning. MLflow 3.0 model registry with staging-to-production promotion gates. Active learning closed loop — when a clinician corrects a prediction, the corrected sample automatically queues for LoRA fine-tuning, the fine-tuned model registers in MLflow, and promotion is gated on F1 and AUC thresholds.”

“Drift detection combining PSI, KS-test, NannyML estimated performance monitoring, and Evidently multivariate analysis — with webhook alerts. Ray Serve with dynamic batching for 16 concurrent images. Circuit breakers on every external service. Immutable audit logs with SHA-256 hash chaining — EU AI Act Article 12 compliant. ONNX and INT8 edge deployment endpoints with output parity validation to four decimal places.”

“188 automated tests. Every feature is opt-in via environment variables. Every feature degrades gracefully when its infrastructure is unavailable. This is how you build medical AI that hospitals will actually deploy.”

5. Q&A Preparation: Top 25 Questions & Answers

Technical

Q: How does this compare to Google’s ARDA system?

“Google ARDA screens for one disease — diabetic retinopathy — and isn’t commercially available after ten years of development. We screen for 45 diseases with a clinical knowledge graph, five explainability methods, and a full production MLOps stack. Google published papers. We built the product.”

Q: What’s your training data? How do you handle class imbalance?

“We train on the RFMiD dataset — 3,200 expert-annotated fundus images across 45 disease labels. Class imbalance is severe — some diseases appear in less than 1% of images. We use asymmetric loss functions, MixUp/CutMix augmentation, and our clinical knowledge graph acts as an implicit regularizer by constraining predictions to clinically plausible combinations.”

Q: 45 diseases sounds like you’re overfitting. What’s your real-world accuracy?

“Valid concern. We’re launching clinical claims with the top 10 diseases and expanding as prospective studies confirm performance. The 45-disease architecture is the platform — the clinical claims follow the evidence. That’s the responsible approach for medical AI.”

Q: Can this run without internet?

“Yes. We have dedicated edge inference endpoints — ONNX Runtime, Core ML for Apple devices, and INT8 quantized models. The model runs locally in Docker, results sync when connectivity is available. The full system runs on CPU for resource-constrained settings.”

Q: How is this different from just fine-tuning a Vision Transformer?

“A fine-tuned ViT gives you probabilities. Our system gives you clinical reasoning. The Knowledge Graph encodes that Diabetic Retinopathy co-occurs with Macular Edema, that Glaucoma has severity stages, that Central RVO is an emergency. On top of that, Claude AI acts as a clinical reasoning agent with a structured six-node pipeline. A ViT doesn’t know any of this.”

Q: What happens when your AI agent (Claude) is down?

“The system has four degradation levels. Full mode uses the Claude agentic pipeline. If Claude is unavailable, circuit breakers detect the failure and the system falls to rule-based clinical reasoning with knowledge graph inference — still clinically valid. If even that fails, raw model predictions with referral priority are still served. The patient is never left without an answer. This is all automated — no human intervention needed.”

Q: How do you monitor the model in production?

“Three layers. First, OpenTelemetry distributed tracing — every prediction generates spans for inference, knowledge graph reasoning, and explainability, visible in Jaeger in real time. Second, drift detection running PSI and KS-test on every 100th prediction, with optional NannyML and Evidently for multivariate analysis. Third, an active learning loop — when drift is detected or clinicians disagree with predictions, the system automatically queues corrected samples for LoRA fine-tuning.”

Clinical / Regulatory

Q: Is this FDA cleared?

“Not yet — we’re pursuing a De Novo pathway because no predicate exists for 45-disease multi-label screening. That means we become the predicate — a regulatory moat. FDA Pre-Sub meeting targeted Q3 2026, CE Mark submission Q4 2026. We’re launching in East Africa first, where registration is faster and the need is most acute.”

Q: What about liability if the AI misses something?

“RetinalAI is a clinical decision support system — a triage and screening aid — not an autonomous diagnostic. The ophthalmologist makes the final diagnosis. Our referral priority ranking (Emergency / Urgent / Routine) ensures high-risk cases are escalated to specialists. And our human-in-the-loop review system catches borderline cases — a clinician reviews any prediction below our confidence threshold.”

Q: How do you handle different cameras and image quality?

“Our fundus validation gate — a fusion of statistical checks and a learned MobileNetV3 classifier — runs in under 12 milliseconds and rejects non-fundus images before inference. If image quality is insufficient, the system asks for re-capture. We’ve tested across multiple camera types.”

Q: What’s your plan for post-market surveillance?

“It’s built in, not bolted on. Every prediction is logged with timestamps, confidence scores, and model versions in an immutable audit trail with SHA-256 hash chaining. Drift detection monitors for distribution shifts in real time. The fairness dashboard tracks performance across demographic groups. Active learning ensures the model improves from clinical feedback. This isn’t a plan — it’s running code with 188 tests.”

Q: How do you ensure fairness across populations?

“Our governance API includes a fairness dashboard endpoint that breaks down model performance by age group, sex, ethnicity, camera device, and geography. Our bias auditor runs automated evaluations. Our clinical knowledge graph includes Uganda-specific prevalence data — not just Western epidemiology. And our active learning loop means the model continuously adapts to the populations it actually serves.”

Business / Market

Q: Why not just sell to the US first? That’s where the money is.

“The US is the most expensive market to enter — FDA clearance costs \$200-350K, the hospital sales cycle is 6-12 months, and you need clinical evidence we don’t have yet. East Africa is where we build clinical evidence with real patients, refine the product, and generate first revenue — all at 1/10th the cost. By the time we enter the US, we have prospective data, CE Mark, and paying customers. That’s a much stronger position.”

Q: How do you compete with incumbents like IDx-DR?

"We don't compete — we complement. IDx-DR screens for one disease at \$40/test. We screen for 45 at \$0.50. They serve US retinal specialists with Topcon cameras. We serve East African primary care clinics with any fundus camera. The overlap is minimal today."

Q: What if Google or a big tech company enters this space seriously?

"Google has been in ophthalmic AI since 2016 and hasn't launched a commercial product. Building the model is 20% of the problem. Regulatory clearance, clinical validation, hospital integration, clinical knowledge graph calibration for specific populations — that's the other 80%. Our LMIC-first strategy gives us clinical data and real-world deployment experience that big tech cannot replicate from Mountain View."

Q: Your revenue projections assume fast clinic adoption. What's the actual sales cycle?

"For specialty clinics: 2-3 months. For hospital networks: 4-6 months. For government programs: 6-12 months. Our Year 1 projections are conservative — 5 enterprise clients, 30 clinics, 2 government programs."

Q: What's your burn rate and runway?

"Post-Seed at \$3M: approximately \$150K/month. That gives us 20 months of runway — enough to reach CE Mark, first revenue, and Series A milestones."

Impact / Vision

Q: What's the long-term vision beyond retinal screening?

"The retina is the only place in the body where you can non-invasively photograph blood vessels and neural tissue. We're starting with eye diseases, but the same image can screen for diabetes, hypertension, cardiovascular risk, and neurological conditions. Our multi-modal fusion architecture is already built to accept OCT scans and patient metadata alongside fundus images. Our federated learning client is ready for cross-hospital training without sharing patient data. The long-term vision is the AI-powered screening gateway for preventive healthcare."

Q: How do you measure impact beyond revenue?

"Patients screened, diseases detected, referrals generated, and sight-threatening conditions caught early. We track time-from-screening-to-treatment as our north star clinical outcome metric. Every metric is logged in our immutable audit trail."

6. Stage Presence & Delivery Tips

Voice & Pacing

- **Slow down for numbers:** “forty-five diseases... two hundred milliseconds... fifty cents per scan” — let them land
- **Speed up for technical details:** Architecture names, MLOps features — confidence, not lecture
- **Pause after the hook:** Let the problem statement sit for 2 seconds before offering the solution
- **End strong:** The last sentence should be a declarative statement, not a question. Drop your pitch, look at the audience

Body Language

- **Stand to the side of the screen,** not in front of it — the demo is the star
- **Point at specific UI elements** when you mention them — guide their eyes
- **Make eye contact with judges/investors** during the problem and close sections
- **Hands visible** — no pockets, no behind-the-back. Open gestures build trust

Demo Execution

- **Pre-position the image** in a visible folder on the desktop
- **Use drag-and-drop,** not file picker — faster and more natural
- **Click deliberately** — rushed clicks make people nervous
- **Flash the Jaeger trace** — even 3 seconds of seeing distributed traces flowing in real time communicates production readiness better than any slide
- **If something loads slowly:** narrate what’s happening rather than standing in silence
- **If the demo breaks:** switch to backup recording with one calm acknowledgment

Common Mistakes to Avoid

1. **Don’t read from slides or notes** — know your pitch cold
 2. **Don’t apologize for what the product doesn’t do yet** — focus on what it does
 3. **Don’t use jargon without context** — say “visual explainability heatmaps” before “GradCAM”
 4. **Don’t rush the demo** — the live demo is your strongest asset
 5. **Don’t end with “any questions?”** — end with your impact statement
 6. **Don’t skip the production stack** — the difference between “I trained a model” and “I built a medical AI platform” is what wins. OpenTelemetry, MLflow, active learning, circuit breakers — these are the words that separate you from every other team
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7. Hackathon Judging Criteria Alignment

Criteria	How RetinalAI Scores	What to Emphasize
Innovation	45-disease multi-label with clinical knowledge graph + agentic AI (Claude + LangGraph)	“No existing system does this — knowledge graph + agentic reasoning is our innovation”
Technical Complexity	RETFound ViT-L + LoRA, 4 GNN architectures, LangGraph 6-node pipeline, OpenTelemetry, MLflow, Ray Serve, 188 tests	Show the architecture briefly, mention the production stack
Impact / Usefulness	2.2B visually impaired, LMIC specialist shortage, WHO screening mandate	Lead with the problem. Make it personal with Grace’s story
Completeness	Full-stack: training, evaluation, backend (10 routers, 47 endpoints), frontend, observability, governance, edge deployment	“This is not a notebook — it’s a production system with 188 tests”
Presentation Quality	Live demo with real predictions + Jaeger traces + governance API	The demo IS the presentation. Flash Jaeger to show production depth
Scalability	Ray Serve dynamic batching, ONNX/INT8 edge, canary releases, circuit breakers, Kubernetes manifests	“Auto-scales from a Raspberry Pi to a GPU cluster”

8. Pitch Deck Slide Outline (If Slides Required)

Slide	Content	5 Questions	Time
1. Title	RetinalAI: “45 diseases. 200ms. 50 cents.”	—	5s
2. The Pain	Grace’s story + stats: 2.2B impaired, 1 ophthalmologist per 1M	Q1 PROBLEM	20s
3. Who Suffers	Buyer personas: clinics, hospitals, ministries	Q2 WHO	15s
4. Solution	One-line: AI screening with clinical reasoning + explainability	Q3 SOLUTION	10s
5. LIVE DEMO	Switch to browser — the core of the pitch	Q3 SOLUTION	60-120s
6. How It Works	Architecture: Image -> Gate -> Model -> KG -> Agent -> Report	Q3 SOLUTION	15s
7. Production Stack	OpenTelemetry + MLflow + Active Learning + Edge + Audit	Q4 WHY YOU	20s
8. Moat	Clinical Knowledge Graph + agentic AI + EU AI Act compliance	Q4 WHY NOW	15s
9. Competitive	2x2 matrix: # diseases vs. price per scan (we’re top-left)	Q4 WHY YOU	15s
10. Market	\$5.4B TAM, 31% CAGR, go-to-market phases	Q2 WHO	20s
11. Business Model	Tier table: per-scan, SaaS, enterprise, OEM. 76% gross margin	Q2 WHO	15s
12. Traction	CE Mark Q4 2026, FDA Q3 2027, Mulago pilot, 188 tests	Q4 WHY NOW	15s
13. Team	Founder backgrounds + key hires planned	Q4 WHY YOU	10s
14. The Ask	“\$2-4M Seed to reach CE Mark + first revenue”	Q5 WHAT YOU WANT	15s
15. Close	“One photograph. 45 diseases. 200 milliseconds. Early detection saves sight.”	—	10s

9. Audience-Specific Adaptations

For Technical Judges (Hackathon)

- Lead with the architecture: “RETFound ViT-Large with LoRA adapters, four GNN architectures, LangGraph agentic pipeline”

- Show code quality: “188 tests, OpenTelemetry distributed tracing, MLflow model registry”
- Emphasize production depth: “Circuit breakers, graceful degradation through four levels, immutable audit logs”
- Flash the Jaeger trace: nothing says “production-ready” like seeing distributed spans flow in real time
- Use precise numbers: “144 disease relationships”, “31 event types in the event bus”, “47 API endpoints”

For Business Judges (Hackathon)

- Lead with market size: “\$5.4 billion ophthalmic AI market, 31% CAGR”
- Emphasize unit economics: “\$0.02 compute cost, \$0.50 price, 76% gross margin”
- Show the competitive moat: “45 diseases vs. competitors’ 1-3, plus knowledge graph + regulatory compliance”
- Mention the timing: “WHO mandated universal eye screening in 2025. EU AI Act enforcement starts 2026. We’re ready.”

For Investors (Pitch)

- Lead with the problem scale and urgency (Q1+Q2 first)
- Focus on defensibility: knowledge graph + agentic AI + regulatory moat + LMIC-first data advantage (Q4)
- Give clear financials: “\$1.76M Year 1 ARR, break-even month 18”
- End with a specific ask and specific milestones (Q5 — always)

For Clinical Audience (Conference/Hospital Demo)

- Lead with patient outcome: “Time-from-screening-to-treatment reduction”
- Show the referral priority system and human-in-the-loop review in detail
- Deep-dive into explainability: “Five methods — GradCAM, LIME, SHAP, Integrated Gradients, ELI5”
- Emphasize the active learning loop: “When you correct a prediction, the model learns from it”
- Show the fairness dashboard: “Performance tracked across age, sex, ethnicity, camera, geography”

For Government / Public Health Buyers

- Lead with population impact: “Screen every diabetic citizen for under \$0.50”
- Emphasize offline capability: “ONNX and INT8 edge deployment — runs without internet”
- Show the governance module: immutable audit trail with SHA-256 integrity, model cards, fairness evaluation

- Mention data sovereignty: “On-premise deployment with federated learning — patient data never leaves your country”
 - Speak to WHO screening guidelines and SDG targets
-

10. Post-Pitch Follow-Up Template

For Investors (Send Within 24 Hours)

Subject: RetinalAI – Follow-up from [Event Name]

Hi [Name],

Thank you for your time at [Event]. As discussed, RetinalAI screens for 45 retinal diseases in <200ms at \$0.50/scan – with clinical reasoning, five explainability methods, and a full 2026 production stack (OpenTelemetry, MLflow, active learning, edge deployment, EU AI Act-compliant audit trail).

Key links:

- Live demo: [URL]
- GitHub: [URL]
- Implementation Roadmap: [URL]
- Commercialization Strategy: [URL]

We're raising a \$2-4M Seed round. I'd welcome 30 minutes to walk through our clinical validation plan and regulatory pathway in detail.

Available [suggest 3 specific times].

Best,

[Your name]

For Potential Partners (Camera OEMs, Hospitals)

Subject: RetinalAI – AI screening integration for [Partner's Product/Hospital]

Hi [Name],

Great meeting you at [Event]. RetinalAI's API can integrate directly with [their camera/EHR system] to add 45-disease AI screening with clinical reasoning and referral prioritization.

Integration options:

- REST API (47 endpoints, OpenAPI documented)
- Docker deployment (GPU or CPU)
- ONNX / Core ML / INT8 edge models for embedded deployment
- Kubernetes-ready with mTLS between services

Compliance: EU AI Act compliant, CE Mark pathway Q4 2026.

Cost: Sub-\$1/scan at volume. Federated learning available for on-premise training without sharing patient data.

I'd love to explore a pilot program. Would [suggest time] work for a technical discussion?

Best,

[Your name]

11. Quick Reference: The 5 Questions at a Glance

Use this as a mental checklist before stepping on stage. If you've answered all five clearly, you've given a complete pitch.

Q1. PROBLEM — Make them feel the pain

Uganda has 60 ophthalmologists for 48 million people — half based in Kampala. 84% of Ugandans are rural with no specialist access. At Mulago Hospital, 19.5% of diabetic patients have retinal disease, 85.7% of cases are sight-threatening. Health spending is \$23/person/year — a quarter of WHO minimum. Current AI tools detect 1-3 diseases at \$25-40/test. Patients go blind waiting.

Q2. WHO — Define the customer

Uganda's 6,000+ health centers with zero eye screening capability. The Ministry of Health (just doubled budget to \$1.5B but can't hire enough specialists). Mulago and the 14 regional referral hospitals. Every Health Centre III/IV across 135 districts seeing diabetic patients but unable to screen their eyes. Ophthalmic Clinical Officers who need AI-assisted triage.

Q3. SOLUTION — 30-second explanation

RetinalAI takes a single photograph of the back of your eye, checks it's a valid retinal image, then screens for 45 diseases in under a fifth of a second — at under UGX 2,000. It shows the clinician exactly why each disease was flagged with a heatmap. It prioritizes which patients need urgent referral to Mulago or regional hospitals. It learns from every correction a doctor makes. It works offline on a laptop — no internet needed. And every result is logged in a tamper-proof audit trail for the MoH.

Q4. WHY YOU / WHY NOW

Why us: We built the only system combining a clinical knowledge graph calibrated for Uganda's disease epidemiology (144 disease relationships), an agentic AI pipeline (Claude + LangGraph), and a production MLOps stack — OpenTelemetry, MLflow, active learning, drift detection, edge deployment, immutable audit logs. 188 tests passing. No competitor has this.

Why now: Uganda launched its National Digital Health Strategic Plan in 2023 and doubled the health budget to \$1.5B in FY 2025/26. WHO mandated universal eye screening in 2025. EU AI Act enforcement starts 2026. The Uganda National Digital Health Conference 2025 called for AI-powered healthcare innovation. The infrastructure moment is now — and there are zero AI eye screening competitors in East Africa.

Q5. THE ASK

\$2-4M Seed round. Use of funds: 40% clinical validation at Mulago Hospital + Makerere partnership + CE Mark, 30% engineering + edge deployment, 20% first three hires (VP Regulatory, Clinical Director, ML Engineer), 10% reserve. Three milestones before Series A: clinical study published with Makerere, first MoH Uganda contract, CE Mark granted. 20-month runway at \$150K/month burn.

12. The Definitive 3-Minute Pitch — Jobs x Peterson Edition

The philosophy: Steve Jobs made you *desire* the product before you understood it — he revealed, he didn't explain. Jordan Peterson makes you feel the *moral weight* of inaction — he doesn't ask you to care, he makes it impossible not to. This pitch fuses both: unveil the product like it's inevitable, frame the mission like it's a moral obligation.

Print this. Read it aloud three times before you sleep the night before. Then once more, standing, ten minutes before you go on.

BEFORE YOU SPEAK

- Dashboard loaded. Image ready. Jaeger in a hidden tab.
 - Stand still. Center stage or slightly left. Hands at your sides.
 - Do NOT touch the computer yet. The first 50 seconds are you and the audience. Nothing else.
 - Breathe. You are about to tell them something that matters.
-

ACT I: THE MORAL WEIGHT

[0:00 - 0:15] THE SILENCE AND THE NUMBER

[Stand completely still. No slides. No screen. Just you and eye contact. Let the silence build for 2 full seconds before you speak. This is the Peterson move — the room should feel like something important is about to be said.]

"Sixty."

[Pause. Two seconds. Let them wonder.]

"That's the number of eye doctors in Uganda. Sixty ophthalmologists... for forty-eight million people."

[Slow, measured, like you're weighing each word.]

"Half of them work in the capital. If you're in Gulu, in Arua, in Kabale — the nearest person who can look into your eyes and tell you whether you're going blind... is a full day's journey away. If you can afford the trip at all."

[0:15 - 0:45] GRACE'S STORY — Make it unbearable NOT to act

[This is Peterson-mode: specific, concrete, morally serious. You are not performing sadness. You are stating facts that carry their own weight. Slower than you think you should go.]

"Let me tell you about a woman named Grace. She's fifty-four. She's diabetic. She lives near a health centre in Gulu District."

"Grace has been losing her peripheral vision for two years. She knows something is wrong. But the nearest ophthalmologist is three hundred and forty kilometers away. A screening costs more than she earns in a month."

[Pause. Lower your voice slightly.]

"Uganda spends twenty-three dollars per person per year on all of healthcare. That's a quarter of the WHO minimum. Grace doesn't have twenty-three dollars to spare on her eyes."

"She waits. And waits."

[Beat.]

"By the time she's seen — advanced glaucoma. Irreversible."

[One more beat. Then, with quiet conviction — not anger, not pity, but clarity:]

"At Mulago Hospital — Uganda's largest — researchers screened diabetic patients. One in five already had retinal disease. And eighty-six percent of those cases... were sight-threatening."

"These are people who went blind because nobody screened them in time. Not because the technology didn't exist. Because it wasn't there."

ACT II: THE REVEAL

[0:45 - 0:55] THE PIVOT — Jobs-style product reveal

[Now your energy shifts. You've established the weight. The audience is uncomfortable. Good. This is where Jobs takes over — the pivot from pain to possibility should feel like a door opening. Turn toward the screen. Your voice gains warmth and certainty.]

"We decided that wasn't acceptable."

[Small pause.]

"So we built something."

[Turn to face the screen. Click to the RetinalAI dashboard. It should already be loaded and beautiful.]

"This is RetinalAI."

[0:55 - 1:55] THE DEMO — Let the product be the hero

[Jobs never over-explained. He showed. He pointed. He let the audience's eyes do the work. Every click is deliberate. Every pause lets the result speak.]

"I'm going to upload a photograph. One photograph — taken with a five-hundred-dollar portable camera. The kind any Health Centre III in Uganda could afford."

[Drag the image. Drop it. Click 'Analyze Image.' While it processes — narrate calmly, like you've seen this a thousand times:]

"First, a safety gate — a neural network fused with statistical validation — checks in twelve milliseconds that this is actually a retinal image. Not a landscape. Not a selfie. A retina."

[Results appear. Point.]

"Two hundred milliseconds."

[Let that number breathe.]

"Forty-five diseases — screened simultaneously. Eleven conditions detected. Referral priority: URGENT. Send this patient to the regional hospital."

[Point at a specific disease on screen.]

"Now look at this. The system didn't just flag diseases in isolation. It reasoned. Our Clinical Knowledge Graph — one hundred and forty-four disease relationships from peer-reviewed ophthalmology literature, calibrated for Uganda's epidemiology — noticed that Diabetic Retinopathy was present and automatically elevated the probability of Cystoid Macular Edema. Because they co-occur."

[Turn to the audience for one sentence:]

"That's not pattern matching. That's differential diagnosis. The kind that takes a specialist years to learn."

[Click GradCAM tab. Point at the heatmap.]

“And this — this is why a doctor will trust it. A heatmap showing exactly which regions of the retina drove the prediction. The ophthalmic clinical officer in Soroti sees the evidence. Confirms it, or corrects it.”

[This next line — deliver it like a Jobs “one more thing” — almost casual:]

“And when they correct it... the model learns. Automatically. It gets better from every scan, at every clinic, across every district.”

[Quick flash to Jaeger — 2 seconds.]

“Every prediction — traced. Every result — logged in a tamper-proof audit trail. Because if the Ministry of Health can’t trust the system, none of this matters.”

ACT III: THE INEVITABILITY

[1:55 - 2:35] WHY THIS, WHY NOW, WHY US — Convergence argument

[Step back from the screen. Face the audience. This is where you synthesize. Peterson’s mode: you’re not selling — you’re explaining why this is the necessary thing to do, and why the timing is not an accident. Jobs’ mode: you’re describing the future as if it’s already happened.]

“People ask me — why now? Why not five years ago?”

[Tick them off on your fingers — three forces converging:]

“Three things just happened at the same time.”

“One. The WHO mandated universal eye screening in 2025. Every member nation — including Uganda — is now committed.”

“Two. Uganda doubled its health budget to one-point-five billion dollars and launched a National Digital Health Strategic Plan. The government is ready for AI in healthcare. They’ve said so publicly.”

“Three. Foundation models changed what’s possible. We fine-tune a model pretrained on one-point-six million retinal images — with only two-point-four million trainable parameters. A LangGraph agentic pipeline where AI reasons through clinical logic. And it runs offline. On a laptop. In a health centre with no internet.”

[Pause. Then the competitive kill shot — understated, factual:]

“And here’s the thing that matters most: there are zero AI eye screening competitors in East Africa. Zero. The market isn’t crowded — it’s empty.”

[2:35 - 3:00] THE CLOSE — Make it impossible to say no

[This is the most important 25 seconds. Slow down. Every word lands like a stone in still water. You are not asking. You are declaring.]

“One photograph.”

[Beat.]

“Forty-five diseases.”

[Beat.]

“Two hundred milliseconds.”

[Beat.]

“Under two thousand shillings. Less than a boda-boda ride.”

[Now — the scope. Let it expand in their minds:]

“We can put this at every Health Centre IV — in every one of Uganda’s one hundred and thirty-five districts. And then every country in East Africa. And then every country where people go blind waiting for a doctor who never comes.”

[Final line. Drop your voice. Not louder — quieter. The room should lean in.]

“We’re raising a Seed round to validate at Mulago Hospital with Makerere University — and deploy to five regional referral hospitals.”

[Look at one person in the audience. Hold their eyes.]

“RetinalAI. Because early detection... saves sight.”

[Hold for 3 full seconds. Do not move. Do not smile. Do not say ‘thank you’ or ‘any questions.’ Let the silence do the work. They will come to you.]

DELIVERY PHILOSOPHY

What Jobs teaches you: - The product is the hero, not you. Point at the screen. Let it speak. - Never explain what they can see. Narrate what they can’t — the reasoning, the knowledge graph, the learning loop. - “One more thing” energy: the GradCAM reveal, the active learning reveal, the

Jaeger trace — each one is a surprise gift. - Numbers spoken slowly are more powerful than numbers on a slide: “Two. Hundred. Milliseconds.”

What Peterson teaches you: - Specificity is credibility. “Grace, fifty-four, diabetic, Gulu District” hits harder than “patients in Uganda.” - Moral seriousness is not sadness. You are not performing grief. You are stating a fact that carries its own gravity: people are going blind because the system failed them. - The argument should feel *necessary*, not clever. You are not pitching — you are explaining what must be done. - Never apologize. Never hedge. “We decided that wasn’t acceptable” is stronger than “we thought we could help.” - End with a statement, never a question. “Early detection saves sight” — period. The audience decides what to do with that.

The fusion: - Peterson opens the wound (the moral weight of inaction). - Jobs heals it (the product that makes the future inevitable). - The audience leaves feeling that investing, partnering, or supporting RetinalAI is not an *opportunity* — it’s a *responsibility*.

Post-Pitch: First 3 Questions You’ll Get

Question	Answer (15 seconds — delivered with the same quiet confidence)
“What’s your accuracy?”	“We’re launching clinical claims on the top 10 diseases first. Expanding as prospective studies at Mulago confirm performance. The forty-five-disease architecture is the platform. Clinical claims follow the evidence. That’s the responsible way to do this.”
“Can it really work offline?”	“Yes. ONNX and INT8 edge models. Runs on a laptop CPU. No internet. Results sync when connectivity returns. We designed for the places that need it most — not the places with the best WiFi.”
“How do you make money at fifty cents?”	“Compute cost: two cents. The rest is margin. Screen Uganda’s one-point-five million diabetic patients — that’s seven hundred fifty thousand dollars in annual revenue. From one country. One disease. Government contracts, SaaS for clinics, OEM licenses for camera manufacturers. This scales.”

The Numbers That Win (Memorize These — Speak Them Slowly)

What	Number	How to say it
Ophthalmologists in Uganda	60	"Sixty. For forty-eight million." (<i>let it land</i>)
DR at Mulago	19.5%	"One in five." (<i>not "nineteen point five percent"</i>)
Sight-threatening	85.7%	"Eighty-six percent of those — sight-threatening."
Health spend	\$23/year	"Twenty-three dollars a year. For everything."
Our cost	UGX 2,000	"Under two thousand shillings. A boda-boda ride."
Diseases	45	"Forty-five. Competitors do three."
Speed	200ms	"A fifth of a second." (<i>snap fingers</i>)
KG relationships	144	"One hundred forty-four peer-reviewed relationships."
Tests	188	"One hundred eighty-eight automated tests. This is not a demo."
Budget	2x	"Uganda just doubled the health budget."
Competitors in EA	0	"Zero. The market isn't crowded. It's empty."

This guide is a living document. Update after every pitch with lessons learned. What resonated? What got blank stares? What questions came up that you didn't anticipate? The pitch gets better with every iteration.